9-2-26 EX:8 NMAP DISCOVER LIVE HOSTS USING NMAP SCANS

AIM:

To discover live hosts on a network using Nmap host-discovery scans ARP scan, ICMP scan and TCP/UDP ping scan - and to understand which protocol operates at which layer of TCP/IP model and when each one applies.

N-MAP:

N-Map ("Network Mapper") is a free open source network scanning tool created by Gordon Lyon and released in 1997. It is industry standard tool for mapping networks, discovering live hosts and identify services running on them.

PROCEDURE:

1. Log into TryHackMe and open NMap Live Host Discovery" system (room).
2. Work through tasks 1-3 and answer Simulator based question.
3. Deploy Lab Network in task 4 and start Attackbox, note target subnet / connection - ID.
4. Run an ARP scan: sudo nmap -PE -sn <subnet>/24 record how many hosts responds.
5. Run an ICMP echo scan: sudo nmap -PE -sn <subnet>/24 compare result with ARP scan.
6. Run ICMP timestamp (-PP) and ICMP Address-Mask scans, note which over ones are filtered.
7. Run TCP SYNC (-PS), TCP ACK (-PA) & UDP (-PU) Ping scans and compare this results and privilege requirements.

8. Optionally capture Nmap traffic using tcpdump or wireshark on the tun0 interface

9. Run nmap -sL <subnet>/24 for target listing and nmap -R <subnet>/24 for reverse-DNS lookup.

10. Complete knowledge check questions (Task 5.2) and proceed to Nmap based port scan room.

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RESULT:

Thus Nmap host discovery scans were performed ~~unsuccessfully~~ successfully using TCP SYN, TCP ACK and UDP ping techniques and the scan results and network traffic were observed.